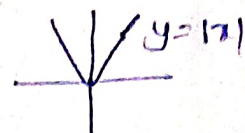
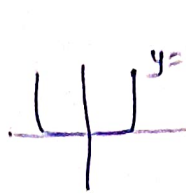
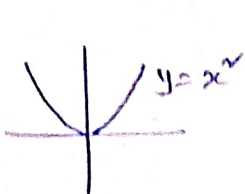
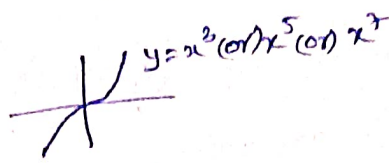
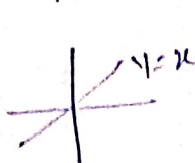


Calculus by Venkata Rao Sir

Even fun : $f(-x) = f(x)$
(Sym about y-axis)



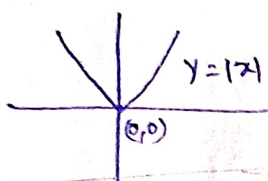
odd fun $f(x) = -f(x)$



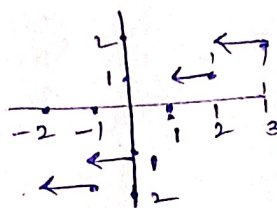
Sym about origin
ie sym in opp quadrants

Modulus fun

$$y = f(x) = |x| = \begin{cases} x & x \geq 0 \\ -x & x < 0 \\ 0 & x = 0 \end{cases}$$



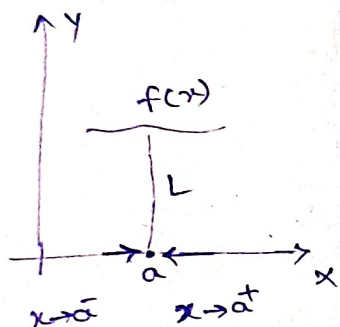
Step fun or Greatest Integer fun



If $0 \leq x < 1$ then $[x] = 0$

$2 \leq x < 3$ then $[x] = 2$

Limit



$\lim_{x \rightarrow a} f(x)$ exists if

- $\lim_{x \rightarrow a^-} f(x) = \text{finite}$
- $\lim_{x \rightarrow a^+} f(x) = \text{finite}$

$$3. \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$$

$$\lim_{x \rightarrow a} x = a$$

Indeterminate forms

$$\frac{0}{0}, \frac{\infty}{\infty}$$

Apply L-Hospital
Rule

$$x \cdot \infty, \infty - \infty, 0 \cdot \infty, \frac{0}{\infty}, \frac{\infty}{0}, 1^\infty$$

Reduce to $\frac{0}{0}$ (or) $\frac{\infty}{\infty}$ form

$$e^\infty = \infty$$

$$e^\infty = \infty$$

$$e^\infty = \frac{1}{e^\infty} = \frac{1}{\infty} = 0$$

$$\ln(1) = 0$$

$$\ln(e) = 1$$

$$\ln(0) = -\infty$$

$$\ln(\infty) = \infty$$

$$1) \lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{x}{\sin x} = 1$$

$$5) \lim_{x \rightarrow 0} \frac{\tan ax}{x} = a$$

$$2) \lim_{x \rightarrow 0} \frac{\sin ax}{x} = a$$

$$6) \lim_{x \rightarrow 0} \frac{\tan ax}{\tan bx} = \frac{a}{b}$$

$$3) \lim_{x \rightarrow 0} \frac{\sin ax}{\sin bx} = \frac{a}{b}$$

$$7) \lim_{x \rightarrow 0} x^n \cdot \log x = 0$$

$$4) \lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{x}{\tan x} = 1$$

$$8) \lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^x = e^a$$

$$9) \lim_{x \rightarrow \infty} (1+ax)^{\frac{1}{x}} = e^a$$

$$\text{Ex: } \lim_{x \rightarrow 0} \frac{|x|}{x}$$

$$\text{Ex: } |x| = \begin{cases} x & x > 0 \\ -x & x < 0 \\ 0 & x = 0 \end{cases}$$

$$\lim_{x \rightarrow 0^-} \frac{|x|}{x} = \lim_{x \rightarrow 0^-} \frac{-x}{x} = \lim_{x \rightarrow 0^-} -1 = -1$$

$$\lim_{x \rightarrow 0^+} \frac{|x|}{x} = \lim_{x \rightarrow 0^+} \frac{x}{x} = \lim_{x \rightarrow 0^+} 1 = 1$$

$$\lim_{x \rightarrow 0^-} \neq \lim_{x \rightarrow 0^+} \text{ So does not exist}$$

$$\frac{0}{0} \text{ indeterminate}$$

$$\text{Ex: } \lim_{x \rightarrow 1} \left[\frac{x^2 - 2ax + b}{x-1} \right] = 8 \text{ then } (a-b) \text{ is}$$

$$\text{Sol: } \frac{1-2a+b}{0} = 8 \Rightarrow 1-2a+b=0 \Rightarrow \boxed{b=2a-1}$$

$$\lim_{x \rightarrow 1} \left[\frac{x^2 - 2ax + 2a - 1}{x-1} \right] = 8$$

App L'H rule

$$b = 2a - 1 = 2(-3) - 1$$

$$b = -7$$

$$\lim_{x \rightarrow 1} \frac{2x - 2a}{1} = 8$$

$$a - b = -3 + 7 = 4$$

$$\boxed{a = -3}$$

$$\text{Ex: } \lim_{x \rightarrow 0} f(x)g(x) = 0 \times \infty \rightarrow 0 \times \frac{1}{0} = \frac{0}{0} \Rightarrow \text{Apply L'H rule}$$

$$\rightarrow \frac{1}{\infty} \times \infty = \frac{\infty}{\infty}$$

If $f(x)$ or $g(x)$ is log fun then $\frac{g(x)}{f(x)} \rightarrow \log \text{ fun}$
 $f(x) \rightarrow \text{other fun}$

Ex: $\lim_{x \rightarrow 0} x \ln(\sin x)$ is $(0 \times -\infty)$

Sol: $\lim_{x \rightarrow 0} \frac{\ln \sin x}{\frac{1}{x}} \left(\frac{-\infty}{\infty} \right)$

$\lim_{x \rightarrow 0} \frac{\left(\frac{\cos x}{\sin x} \right)}{-\frac{1}{x^2}} \Rightarrow -\lim_{x \rightarrow 0} x^2 \cot x$

$\Rightarrow -\lim_{x \rightarrow 0} x \left(\frac{x}{\tan x} \right)$

$\Rightarrow -0 \times 1 = 0.$

Model IV $\infty - \infty$ form

$\lim_{x \rightarrow a} [f(x) - g(x)] = \infty - \infty$

Take LCM $\begin{cases} \frac{0}{0} \\ \frac{\infty}{\infty} \end{cases} \Rightarrow \text{Apply L-Hosp rule}$

Model V $0^0, \infty^0, 1^\infty$ form

$\lim_{x \rightarrow a} [f(x)]^{g(x)} = 0^0 \text{ or } \infty^0 \text{ or } 1^\infty$

process: let $y = \lim_{x \rightarrow a} [f(x)]^{g(x)}$

$\ln y = \lim_{x \rightarrow a} \ln f(x) \cdot g(x)$

$\ln y = \lim_{x \rightarrow a} g(x) \ln f(x)$ $0 \times \infty$

$y = e^{\lim_{x \rightarrow a} g(x) \ln f(x)}$ $e^{0 \times \infty}$

apply L-H rule

Ex: $\lim_{x \rightarrow \infty} (1+x^2)^{e^{-x}}$

Sol: $y = e^{\lim_{x \rightarrow \infty} e^{-x} \ln(1+x^2)}$ $e^{0 \times \infty}$

$= e^{\lim_{x \rightarrow \infty} \frac{\ln(1+x^2)}{e^x}}$ $e^{\frac{0}{\infty}}$

$= e^{\lim_{x \rightarrow \infty} \frac{2x}{(1+x^2)e^x}}$ $e^{\frac{\infty}{\infty}}$

$y = e^{\lim_{x \rightarrow \infty} \frac{2}{(1+x^2)e^x + 2xe^x}}$

$y = e^{\frac{2}{\infty}} = e^0 = 1$

Continuity and Differentiability

Continuity

$$\lim_{x \rightarrow \bar{a}} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$$

Left
Continuity

Right
Continuity

Differentiable fun

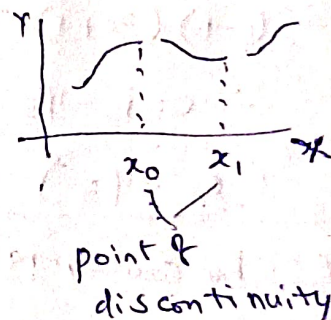
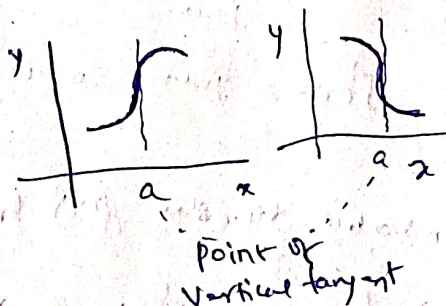
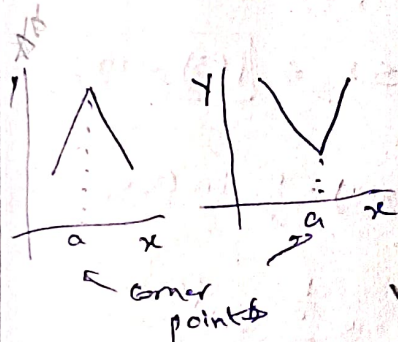
$$f'(a) = f'(a^+)$$

$y = |x|$ — Continuity everywhere except at $x=0$
Differentiable

Fun failed to be differentiable in following cases

1. Corner point (sharp corner edge) : 2. point of vertical tangent

3. point of discontinuity



Ex: $f(x) = \begin{cases} e^x & x < 1 \\ \ln x + ax^2 + bx & x \geq 1 \end{cases}$ find (a, b)

Let $f(x)$ is differentiable at $x=1$

So continuity at $x=1$

$f(x)$ is differentiable for unique

values of a, b

$$e = \ln(1) + a(1)^2 + b(1)$$

$$e = a + b \rightarrow (1)$$

$$a + b = e$$

$$-1 + b = e$$

$$b = e + 1$$

$$f'(x) = \begin{cases} e^x & x < 1 \\ \frac{1}{x} + 2ax + b & x \geq 1 \end{cases}$$

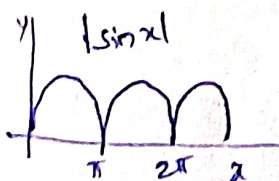
$$e = 1 + 2a + b$$

$$2a + b = e - 1 \rightarrow (2)$$

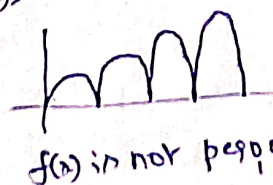
$$2a + b = e - 1 \quad (2) - (1)$$

$$\frac{-a + b = -e}{a = -1}$$

Ex: $f(x) = e^x |\sin x|$



$$f(x) = e^x |\sin x|$$



$f(x)$ is not periodic fun

Mean Value Theorem

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Ex: $f(x) = 1 - x^2 + x^3$ in $(-1, 1)$ Mean then stp

$\Rightarrow f'(x) = -2x + 3x^2$ $\frac{a}{a} \frac{b}{b}$

$$f'(c) = \frac{f(b) - f(a)}{b - a} \Rightarrow -2c + 3c^2 = \frac{f(1) - f(-1)}{1 - (-1)} \Rightarrow 3c^2 - 2c = \frac{1 - (-1)}{1 - (-1)} = 1$$

$$3c^2 - 2c - 1 = 0$$

$$3c^2 - 3c + c - 1 = 0$$

$$3c(c-1) + c - 1 = 0$$

$$(c-1)(3c+1) = 0$$

$$\therefore c = 1, -1/3$$

$$\boxed{c = -1/3}$$

Cauchy's Mean Value Theorem

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

Ex: $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous function on $(-3, 3)$ such that diff^s fun

$\times \{f'(x) \leq 2\}$. If $f(-3) = 7$ then $f(3)$ is —

$\Rightarrow a = -3, b = 3$ By mean value theorem,

$$f'(c) = \frac{f(b) - f(a)}{b - a} \Rightarrow f'(c) = \frac{f(3) - f(-3)}{3 - (-3)} \Rightarrow f'(c) = \frac{f(3) - 7}{6}$$

$$f'(x) \leq 2 \text{ given, so } \frac{f(3) - 7}{6} \leq 2 \Rightarrow \boxed{f(3) \leq 19}$$

Ex: By Cauchy's Mean Value theorem $f(x) = \ln x$ & $g(x) = \frac{1}{x}$ in $(1, e)$

Sol: $f(x) = \ln x$ $g(x) = \frac{1}{x}$

$$f'(x) = \frac{1}{x} \quad g'(x) = -\frac{1}{x^2}$$

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)} \Rightarrow \frac{\frac{1}{c}}{-\frac{1}{c^2}} = \frac{f(e) - f(1)}{g(e) - g(1)}$$

$$\Rightarrow -c = \frac{\ln e - \ln 1}{\frac{1}{e} - 1}$$

$$\boxed{c = \frac{e}{e-1}}$$

Taylor's series

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \frac{(x-a)^4}{4!}f^{(4)}(a) + \dots$$

if $a=0$ expansion about $x=0 \Rightarrow$ Maclaurin's series.

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \frac{x^4}{4!}f^{(4)}(0)$$

if $a=0$, about $x=0$

$$1. \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \quad |x| < \infty$$

$$2. \sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \dots \quad |x| < \infty$$

$$3. \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \quad |x| < \infty$$

$$4. \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \dots \quad |x| < \infty$$

$$5. e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \quad |x| < \infty$$

$$6. e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \quad |x| < \infty$$

$$7. \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \quad |x| < 1$$

$$8. \ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} + \dots \quad |x| < 1$$

$$9. \frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots \quad |x| < 1$$

$$10. \frac{1}{1+x} = (1+x)^{-1} = 1 - x + x^2 - x^3 + \dots \quad |x| < 1$$

$$11. (1-x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \dots \quad |x| < 1$$

$$12. (1+x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \dots \quad |x| < 1$$

Ex: Let $f(x)$ be thrice diff fun on $(-\frac{\pi}{6}, \frac{\pi}{6})$ such that $f'(x) = 1 + [f(x)]^2$. If $f(0) = 1$

then coeff of x^2 in Taylor series expansion about $x=0$ is —

Sol: $f'(x) = 1 + [f(x)]^2$, $f(0) = 1$ | Ex: $e^x + \sin x$ about point $x=\pi$ coeff $(x-\pi)^r$

$$f''(x) = 2f(x)f'(x)$$

$$\text{Sol: } f(x) = e^x + \sin x$$

$$f'(0) = 2f(0)f'(0)$$

$$f'(x) = e^x + \cos x$$

$$f''(0) = 2f(0) \{1 + f(0)^2\}$$

$$f''(x) = e^x - \sin x$$

$$f''(0) = 2(1)(1+1) = 4$$

$$f''(\pi) = e^\pi - \sin \pi = e^\pi$$

$$x^2 \text{ coeff} = \frac{f''(0)}{2!} = \frac{4}{2} = 2$$

$$(x-\pi)^r \text{ coeff} = \frac{f^{(r)}(\pi)}{r!} = \frac{f''(\pi)}{2!} = \frac{1}{2} e^\pi = 0.5 e^\pi$$

Ex: Taylor series $\frac{\sin x}{x-\pi}$ about $x=\pi$ is

$$\text{Sol: } x-\pi = t \Rightarrow x = \pi + t$$

$$\sin x = \sin(\pi + t)$$

$$\sin x = -\sin t$$

$$\text{Now } \frac{\sin x}{x-\pi} = \frac{-\sin t}{t} = -\frac{1}{t} \left\{ t - \frac{t^3}{3!} + \frac{t^5}{5!} - \frac{t^7}{7!} + \dots \right\}$$

$$\Rightarrow -1 + \frac{t^2}{3!} - \frac{t^4}{5!} + \frac{t^6}{7!} - \dots \Rightarrow \frac{\sin x}{x-\pi} = -1 + \frac{(x-\pi)^2}{3!} - \frac{(x-\pi)^4}{5!} + \frac{(x-\pi)^6}{7!} - \dots$$

Partial Differentiation

Total derivative.

If $u = f(x, y)$ where $x = \phi_1(t)$ & $y = \phi_2(t)$ then total derivative w.r.t t

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial y} \frac{dy}{dt}$$

Chain rule

$u = f(x, y, z, t)$ where $x = \phi_1(x, y, z, t)$, $y = \phi_2(x, y, z, t)$, $z = \phi_3(x, y, z, t)$, $t = \phi_4(x, y, z, t)$

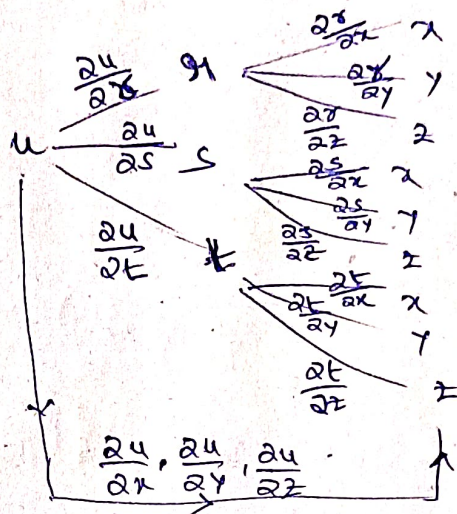
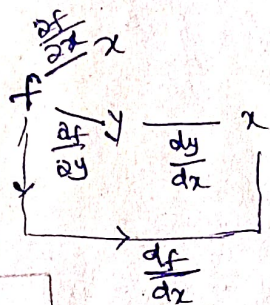
$$\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \frac{dy}{dx} + \frac{\partial u}{\partial z} \frac{dz}{dx} + \frac{\partial u}{\partial t} \frac{dt}{dx}$$

Derivative of Implicit fun

$$\frac{df}{dx} = 0$$

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = \frac{-f_x}{f_y}$$



Ex: $z(x, y) = e^{x-2y}$ $x(t) = e^t$ $y(t) = e^t$

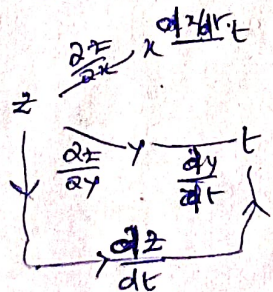
then $\frac{dz}{dt}$ is

Sol:

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

$$\Rightarrow e^{x-2y} \cdot e^t + 2e^{x-2y} \cdot e^t$$

$$\frac{dz}{dt} = 3e^{x-2y}$$



Ex: $u = x \ln(xy)$, $x^3 + y^3 + 3xy = 1$, $\frac{du}{dx}$

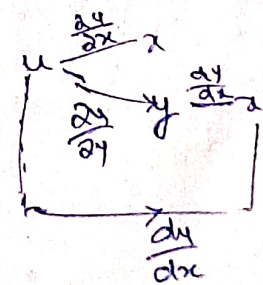
$$\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \frac{dy}{dx}$$

$$\Rightarrow \ln(xy) + x \frac{y}{xy} + x \frac{x}{xy} \frac{dy}{dx}$$

$$\frac{du}{dy} = 1 + \ln(xy) + \frac{x}{y} \frac{dy}{dx} \rightarrow (1)$$

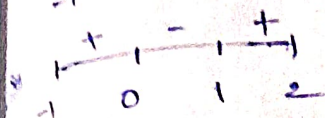
$$\frac{-(x^2y)}{(y^3+x)} = \frac{-(3x^2y+3y)}{(3y^2+3x)} \leftarrow \frac{dy}{dx} = \frac{-f_x}{f_y}$$

$$\frac{du}{dx} = 1 + \ln(xy) - \frac{x(x^2y)}{y^3+x}$$



Definite Integrals

1. $\int_{-1}^2 |x^3 - x| dx$ —



$$x^3 - x = 0$$

$$x(x^2 - 1) = 0$$

$$x = 0, 1, -1$$

2. $\int_{-1}^2 |x^3 - x| dx$

$$\int_{-1}^0 (x^3 - x) dx - \int_0^1 (x^3 - x) dx + \int_1^2 (x^3 - x) dx$$

$$\left[\frac{x^4}{4} - \frac{x^2}{2} \right]_{-1}^0 - \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_0^1 + \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_1^2$$

$$\Rightarrow \left[(0) - \left(\frac{1}{4} - \frac{1}{2} \right) \right] - \left[\frac{1}{4} - \frac{1}{2} - 0 \right] + \left[(4 - 2) - \left(\frac{1}{4} - \frac{1}{2} \right) \right]$$

$$\Rightarrow -\frac{3}{4} + \frac{3}{2} + 2 = \frac{-3 + 6 + 8}{4} = \frac{11}{4}$$

1. $\int_0^{1.5} x[x^2] dx$ —

$$\Rightarrow \int_0^{2.25} t \cdot \frac{1}{2} dt$$

putting $x^2 = t$

$$2x dx = dt$$

$$x dx = \frac{1}{2} dt$$

$$\frac{1}{2} \left\{ \int_0^1 0 dt + \int_1^2 1 dt + \int_2^{2.25} 2 dt \right\}$$

x	t
0	0
1.5	2.25

$$\Rightarrow \frac{1}{2} \left[(t)_1 - 2(t)_2 \right]$$

$$x^2 = 0^2 = 0$$

$$x^2 = 1.5^2 = 2.25$$

$$= \frac{1}{2} \left\{ (2-1) + 2(2.25-2) \right\}$$

$$\int_0^{1.5} x(x^2) dx = 0.75$$

2. $\int_0^{\pi/6} \cos^4(3\theta) \sin^3(6\theta) d\theta$

3. put $3\theta = t$

$$3d\theta = dt$$

$$d\theta = \frac{1}{3} dt$$

θ	t
0	0
$\frac{\pi}{6}$	$\frac{\pi}{2}$

$$\Rightarrow \int \cos^4(t) \sin^3 t \cdot \frac{1}{3} dt$$

$$= \frac{1}{3} \int_0^{\pi/2} \cos^4(t) \{ 2 \sin t \cos t \}^3 dt$$

$$= \frac{8}{3} \int_0^{\pi/2} \sin^3 t \cos^7 t dt$$

$$= \frac{8}{3} \frac{2 \times 6 \times 4 \times 2}{10 \times 8 \times 6 \times 4 \times 2} = \frac{1}{15}$$

Improper Integral

Ex: $\int_{-1}^1 \frac{1}{x^2} dx$ — even fun

$$\Rightarrow 2 \int_0^1 \frac{1}{x^2} dx \Rightarrow 2 \lim_{a \rightarrow 0^+} \int_a^1 \frac{dx}{x^2} = 2 \lim_{a \rightarrow 0^+} \left(-\frac{1}{x} \right)_a^1$$

$$\Rightarrow 2 \lim_{a \rightarrow 0^+} \left(-1 + \frac{1}{a} \right) \Rightarrow 2 \left[-1 + \frac{1}{0} \right] = 2(\infty - 1)$$

(divergent or unbounded) = ∞

Ex: $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$, If $a < b$ the value $\int_{-\infty}^{\infty} e^{-a(x+b)^2} dx$ is

$$\Rightarrow \int_{-\infty}^{\infty} e^{-a(x+b)^2} dx = \int_{-\infty}^{\infty} e^{-[\sqrt{a}(x+b)]^2} dx \xrightarrow{\text{putting}} \sqrt{a}(x+b) = t$$

$$\Rightarrow \int_{-\infty}^{\infty} e^{-t^2} \frac{dt}{\sqrt{a}} \Rightarrow \frac{1}{\sqrt{a}} \sqrt{\pi}$$

x	t
$-\infty$	$-\infty$
∞	∞

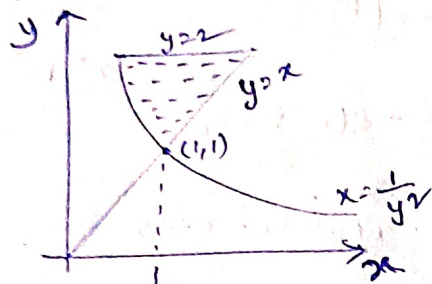
Area

$$\text{Area} = \int_a^b \{ \text{upper curve } y \text{ value} \} - \{ \text{lower curve } y \text{ value} \} dx$$

$$= \int_c^d \{ \text{Right curve } x \text{ value} \} - \{ \text{left curve } x \text{ value} \} dy$$

$$= \iint dx dy$$

1) $y = x^2 + 1$ parabola



$$\text{So Area} = \int_1^2 (y - \frac{1}{y^2}) dy = \left[\frac{y^2}{2} + \frac{1}{y} \right]_1^2$$

$$= (2 + \frac{1}{2}) - (\frac{1}{2} + 1)$$

$$\text{Area} = 1 \text{ sq. unit}$$

$$y = x \Rightarrow x = \frac{1}{y^2}$$

$$y = \frac{1}{y^2}$$

$$y^3 = 1 \Rightarrow y = 1, x = 1$$

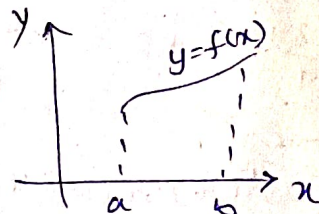
Length of curve

$$y = f(x) \quad x=a, x=b$$

$$L = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$y = f(y), \quad y=c, y=d$$

$$L = \int_c^d \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$



Length of spatial curve in parametric form $x = f_1(t), y = f_2(t), z = f_3(t)$ s.t. $t = a$ to $t = b$

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

Ex: len of curve $y = \ln \sec x$ from $x=0, \frac{\pi}{4}$

$$\text{So } y = \ln(\sec x)$$

$$\frac{dy}{dx} = \frac{\sec x \tan x}{\sec x} = \tan x$$

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \tan^2 x = \sec^2 x$$

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{\sec^2 x} = \sec x$$

$$L = \int_0^{\frac{\pi}{4}} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \Rightarrow \int_0^{\frac{\pi}{4}} \sec x dx \Rightarrow \left[\ln(\sec x + \tan x) \right]_0^{\frac{\pi}{4}}$$

$$\Rightarrow \ln\left(\sec \frac{\pi}{4} + \tan \frac{\pi}{4}\right) - \ln(\sec 0 + \tan 0)$$

$$\Rightarrow \ln(\sqrt{2} + 1) - \ln(1)$$

$$L = \ln(1 + \sqrt{2})$$

Ex: $x(t) = \cos t, y(t) = \sin t, z(t) = \frac{2}{\pi} t, 0 \leq t < \frac{\pi}{2}$ length of curve Spatial curve (3D)

$$\frac{dx}{dt} = -\sin t, \quad \frac{dy}{dt} = \cos t, \quad \frac{dz}{dt} = \frac{2}{\pi}$$

$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} = \sqrt{\sin^2 t + \cos^2 t + \frac{4}{\pi^2}} = \sqrt{1 + \frac{4}{\pi^2}} = 1.85$$

$$L = \int_0^{\frac{\pi}{2}} 1.85 dt = 1.85 \left(\frac{\pi}{2} \right)$$

$$\Rightarrow \frac{\pi}{2} (1.85) = 1.86$$

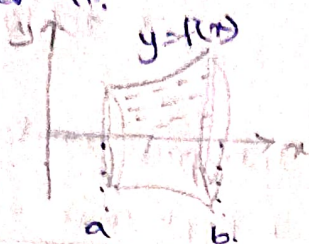
Vol of Solid of Revolution

1. Area bounded by curve

$y=f(x)$, $x=a$, $x=b$ is rev about x-axis

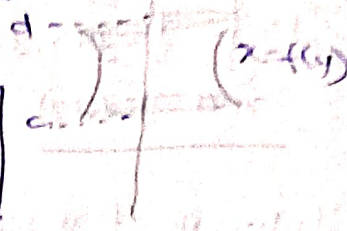
Vol of solid of rev is

$$V = \int_a^b \pi y^2 dx$$



2. $x=f(y)$, $y=c$, $y=d$ about y-axis

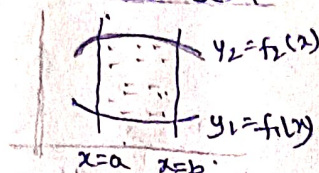
$$V = \int_c^d \pi x^2 dy$$



3. Area bounded by curves $y_1=f_1(x)$, $y_2=f_2(x)$

between $x=a$ & $x=b$ about x-axis then

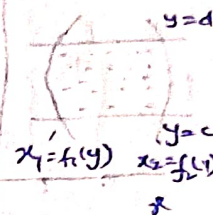
$$V = \int_a^b \pi (y_2^2 - y_1^2) dx$$



$x_1=f_1(y)$, $x_2=f_2(y)$

between $y=c$ & $y=d$

$$V = \int_c^d \pi (x_2^2 - x_1^2) dy$$



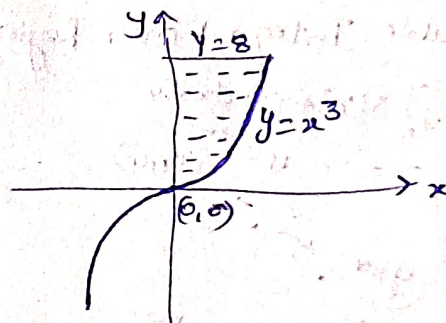
Ex: Vol of solid by rotating the region bounded by $y=x^3$, $y=8$ & $x=0$

about y-axis is

$$Vol = \int_c^d \pi x^2 dy = \int_0^8 \pi y^{\frac{2}{3}} dy$$

$$y = x^3$$

 $x = y^{1/3}$
 $x^2 = y^{2/3}$



$$= \pi \left\{ \frac{y^{5/3}}{5/3} \right\}_0^8 \Rightarrow \frac{3\pi}{5} \left\{ 8^{5/3} - 0^{5/3} \right\}$$

$$\Rightarrow 96 \frac{\pi}{5} \Leftrightarrow \frac{3\pi}{5} \left\{ (2^3)^{5/3} - 0 \right\}$$

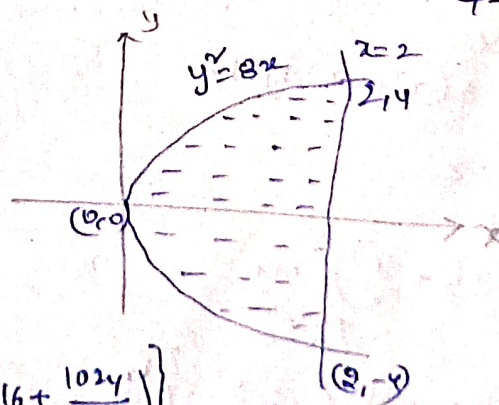
Ex: Vol gen by rev area bounded by parabola $y^2=8x$ & line $x=2$ about y-axis

$$Vol = \int_c^d \pi (x_2^2 - x_1^2) dy$$

$$\Rightarrow \pi \int_{-4}^4 \left\{ 2^2 - \left(\frac{y^2}{8} \right)^2 \right\} dy \Rightarrow \pi \int_{-4}^4 \left(4 - \frac{y^4}{64} \right) dy$$

$$= \pi \left\{ 4y - \frac{y^5}{320} \right\}_{-4}^4 \Rightarrow \pi \left\{ \left(16 - \frac{1024}{320} \right) - \left(-16 + \frac{1024}{320} \right) \right\}$$

$$= 32\pi \left(1 - \frac{1}{5} \right)$$



$$= \frac{128}{5} \pi$$

Multiple Integrals

$$I = \int_{x=a}^b \int_{y=f_1(x)}^{f_2(x)} f(x,y) dy dx$$

$$I = \int_{y=c}^d \int_{x=f_1(y)}^{f_2(y)} f(x,y) dx dy$$

Integrate first w.r.t. y
 x kept const then then
 resulting fun Integrated w.r.t x

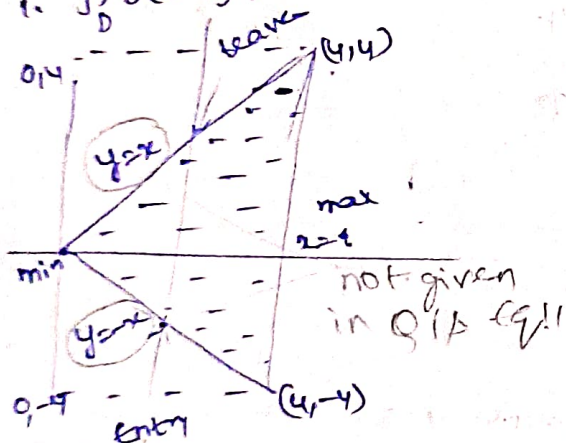
Integrate w.r.t x first where
 y kept const and then resulting fun
 integrate w.r.t y

Ex: $\int_0^{\pi/2} \int_0^{\cos \theta} x \sin \theta dx d\theta$

Sol: $\Rightarrow \int_0^{\pi/2} \sin \theta \left(\frac{x^2}{2} \right)_0^{\cos \theta} d\theta \Rightarrow \frac{1}{2} \int_0^{\pi/2} \sin \theta \cos^2 \theta d\theta$

Double Integral over Region

1. $\iint_D 3(x^2 + y^2) dx dy$



$$\Rightarrow \int_0^4 \int_{-x}^x 3(x^2 + y^2) dy dx$$

$x=0$ $y=x$

$$\Rightarrow 3 \int_0^4 \left(x^2 y + \frac{y^3}{3} \right)_{y=-x}^{y=x} dx$$

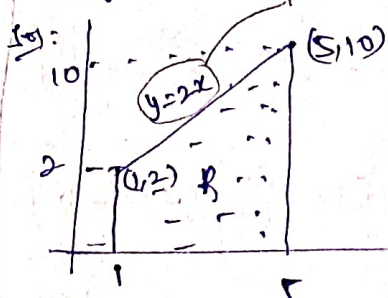
$$\Rightarrow 3 \int_0^4 \left(x^2 \left(\frac{x^3}{3} + \frac{x^3}{3} \right) - \left(-x^2 \frac{x^3}{3} \right) \right) dx$$

$$= 3 \int_0^4 \frac{8}{3} x^3 dx \Rightarrow (8x^4)_0^4$$

$$= 2(4^4) = 512$$

$$\Rightarrow \frac{1}{2} \left\{ \frac{1}{3 \times 1} \right\} = 1/6$$

Q2) $I = c \int \int_R xy^2 dx dy$ $c = 6 \times 10^{-4}$
 region shown in fig



$$I = c \int_{x=1}^5 \int_{y=0}^{2x} xy^2 dy dx \Rightarrow c \int_1^5 x \left(\frac{y^3}{3} \right)_0^{2x} dx$$

$$= \frac{c}{3} \int_1^5 2 \left(\frac{8}{3} x^3 \right) dx \Rightarrow \frac{8c}{9} \int_1^5 x^3 dx$$

$$\Rightarrow \frac{8c}{9} \left(\frac{x^4}{4} \right)_1^5$$

$$\Rightarrow \frac{8c}{9 \times 4} \{ 5^4 - 1 \}$$

$$= \frac{8 \times 6 \times 10^{-4}}{36} (3125 - 1)$$

$$I = 0.95$$

Vol of double Integrals

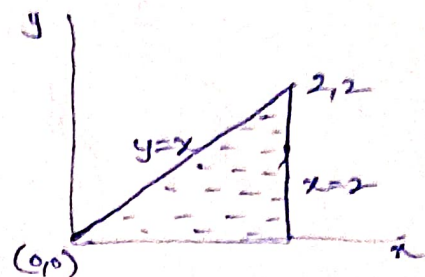
Ex: vol of prism whose base is Δ^e in xy plane bounded by x -axis and the line $y=x$ & $x=2$ and where top is plane $z=5-x-y$ is —

$$\text{Vol} = \iint \Delta \, dxdy \Rightarrow \iint (5-x-y) \, dxdy$$

$$\text{Vol} = \int_{x=0}^2 \int_{y=0}^x (5-x-y) \, dy \, dx$$

$$= \int_0^2 \left(5y - xy - \frac{y^2}{2} \right)_0^x \, dx \Rightarrow \int_0^2 \left(5x - x^2 - \frac{x^2}{2} \right) \, dx \Rightarrow \int_0^2 \left(5x - \frac{3}{2}x^2 \right) \, dx$$

$$= \left[\frac{5x^2}{2} - \frac{3x^3}{2} \right]_0^2 \Rightarrow 10 - 4 = 6$$

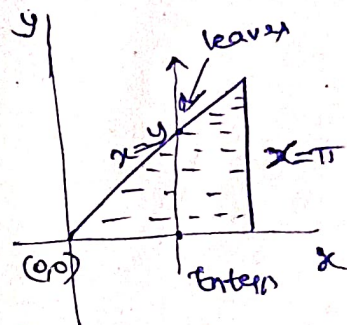


change order

Ex: The value of Integral $\int_0^\pi \int_y^\pi \frac{\sin x}{x} \, dxdy$ is equal to —

$$\text{Sol: } \int_{y=0}^\pi \int_x=y^\pi \frac{\sin x}{x} \, dxdy$$

changing order of Integration



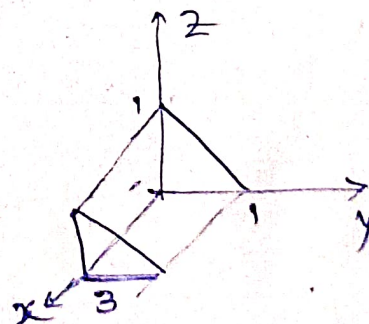
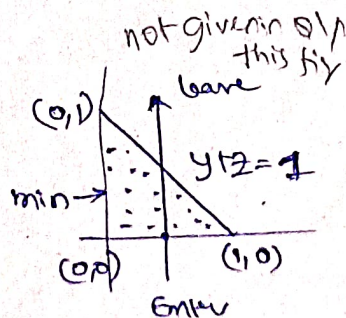
$$\Rightarrow \int_{x=0}^\pi \int_{y=0}^x \frac{\sin x}{x} \, dy \, dx \Rightarrow \int_0^\pi \frac{\sin x}{x} (y)_0^x \, dx$$

$$\Rightarrow \int_0^\pi \frac{\sin x}{x} x \, dx \Rightarrow \left\{ -\cos x \right\}_0^\pi \Rightarrow \left\{ -\cos \pi + \cos 0 \right\} = 2$$

Triple Integrals

$$\text{Ex: } \iiint \Delta \, dxdy \, dz = ?$$

$$\text{Sol: } \int_{y=0}^1 \int_{z=0}^{1-y} \int_{x=0}^3 x \, dxdy \, dz$$



$$= \int_{y=0}^1 \int_{z=0}^{1-y} x (z)_0^{1-y} \, dxdy \Rightarrow \int_{y=0}^1 \int_{z=0}^{1-y} x(1-y) \, dxdy \Rightarrow \int_0^1 \left(\frac{x^2}{2} - \frac{x^2}{2} \right)_0^{1-y} \, dy$$

$$\Rightarrow \int_0^1 \left(\frac{9}{2} - \frac{9}{2}y \right) \, dy \Rightarrow \left[\frac{9}{2}y - \frac{9y^2}{4} \right]_0^1$$

$$= 9/2 - 9/4 = 2.25$$

Vector Calculus

$\phi = \text{fun}(x, y, z)$ scalar

grad $\phi = \nabla \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}$

$\nabla \phi = \frac{\partial \phi}{\partial x} \hat{i} + \frac{\partial \phi}{\partial y} \hat{j} + \frac{\partial \phi}{\partial z} \hat{k}$

unit Normal vector to the surf $\phi(x, y, z) = \frac{\nabla \phi}{|\nabla \phi|}$

Rate of change of scalar fun ϕ in direction of vector $\vec{a}$

Directional derivative of ϕ in direction of $\vec{a} = \nabla \phi \cdot \frac{\vec{a}}{|\vec{a}|}$

Divergence $\text{div}(\vec{F}) = \nabla \cdot \vec{F}$

$\vec{F} = F_1 \hat{i} + F_2 \hat{j} + F_3 \hat{k}$

$\text{div}(\vec{F}) = \frac{\partial}{\partial x}(F_1) + \frac{\partial}{\partial y}(F_2) + \frac{\partial}{\partial z}(F_3)$

$\text{curl}(\vec{F}) = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$

$\text{grad}[f(x)] = f'(x) \hat{i}$
 $\nabla \{f(x)\} = f'(x) \hat{i} + \frac{\partial}{\partial y} f(x) \hat{j}$

$\vec{u} = y\hat{i} + x\hat{j}$
 $\vec{v} = x\hat{i} + y\hat{j}$

$\vec{u} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ y & x & 0 \\ x & y & 0 \end{vmatrix}$

Prob
 $A = 2\hat{j} - 3\hat{k}$
 $B = 2\hat{j} + \hat{k}$
 $C = 2\hat{i} - \hat{j}$
 The value $[\vec{A} \cdot (\vec{B} \times \vec{C}) + 6]$

$\vec{B} \times \vec{C} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 2 & 1 \\ 2 & -1 & 0 \end{vmatrix} = \hat{i}(0-2) - \hat{j}(0-2) + \hat{k}(2-0) = -2\hat{i} + 2\hat{j} + 2\hat{k}$

$\vec{A} \cdot (\vec{B} \times \vec{C}) = (2\hat{j} - 3\hat{k}) \cdot (-2\hat{i} + 2\hat{j} + 2\hat{k}) = 0 + 4 - 6 = -2$

$[\vec{A} \cdot (\vec{B} \times \vec{C}) + 6] = -2 + 6 = 4$

Imp

$\text{div}(\text{curl} \vec{V}) = 0$

∇f is Irrational vector field

$\text{curl}(\nabla f) = 0$ $\vec{V} = 3x\hat{i} + x\hat{k}$

Irrational $\text{curl} \vec{V} = 0$

print much imp
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Solve mo

Vector Integration

Line Integral

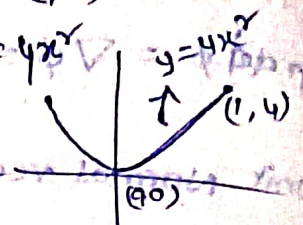
2D path given

work done = $\int_C \vec{F} \cdot d\vec{r} = \int_{(0,0)}^{(1,4)} (\vec{F}_1 + \vec{F}_2) \cdot (dx\vec{i} + dy\vec{j})$

from (0,0) to (1,4)

$\vec{F} = 2xy\vec{i} + 3xy\vec{j}$
 $\vec{F}_1 = 2xy\vec{i}$
 $\vec{F}_2 = 3xy\vec{j}$

along $y = 4x^2$



$\Rightarrow \int_{(0,0)}^{(1,4)} F_1 dx + F_2 dy$

$= \int F_1 dx + F_2 dy$

$dy = 4(2x)dx$

$2xy dx + 3xy dy = \int 2x(4x^2) dx + 3x(4x^2) 8x$

$dy = 8x dx$

3D path given

P(1,1,2) Q(2,2,1)

terms in x including the dy

$\int_C \vec{F} \cdot d\vec{r} = \int_P^Q F_1 dx + F_2 dy + F_3 dz$

limit $\int_{t_1}^{t_2}$ all in terms of t

$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1} = t$

Convert to like this

$x = t+1, y = 2t+1, z = 2-t$

change limits also

$dx = dt, dy = 2dt, dz = -dt$

3D path not given

$\vec{P} = 2\vec{i} - 5\vec{j} + 6\vec{k}$

If $\text{curl}(\vec{P}) =$

$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2 & -5 & 6 \end{vmatrix} = 0$

If in given all are constant
 directly $\text{Curl } \vec{P} = 0$

$A = (6, 1, 3)$
 $6\vec{i} + \vec{j} + 3\vec{k}$
 $B = (4, 3, 2)$
 $4\vec{i} + 3\vec{j} + 2\vec{k}$

$\phi = \int F_1 dx + F_2 dy + F_3 dz$

$\oint \vec{F} \cdot d\vec{r} = \phi(B) - \phi(A)$

$\oint \vec{F} \cdot d\vec{r}$

$0 = 0 + 0 = [0 + (5 \times 8) \cdot \vec{k}]$

$0 = (\nabla \cdot \omega) \text{ with } \dots$

Surf. Integral $\rightarrow$ closed paths

Stokes theorem

$$\oint_C \vec{F} \cdot d\vec{s} = \iint_S (\text{Curl } \vec{F}) \cdot \vec{n} \, dS$$

If $\text{Curl } \vec{F} = 0$ then $\oint_C \vec{F} \cdot d\vec{s} = 0$

already apply
Green theorem

$$\oint_C \vec{F} \cdot d\vec{s} = \iint_D \left(\frac{\partial}{\partial x} F_2 - \frac{\partial}{\partial y} F_1 \right) dx dy$$

$$\oint_C F_1 dx + F_2 dy = \iint_D \left(\frac{\partial}{\partial x} F_2 - \frac{\partial}{\partial y} F_1 \right) dx dy$$

line $\rightarrow$ surf: $\vec{r} = \frac{y}{2}\vec{i} + \frac{y}{2}\vec{j} + x\vec{k}$ around a circle $x^2 + y^2 = 4$
if 16π then $|a|$

$$\vec{F}(x,y) = (2x-ay)\vec{i} + (2y+ax)\vec{j}$$

$$\oint_C \vec{F} \cdot d\vec{s} = \iint_D \left(\frac{\partial}{\partial x} F_2 - \frac{\partial}{\partial y} F_1 \right) dx dy$$

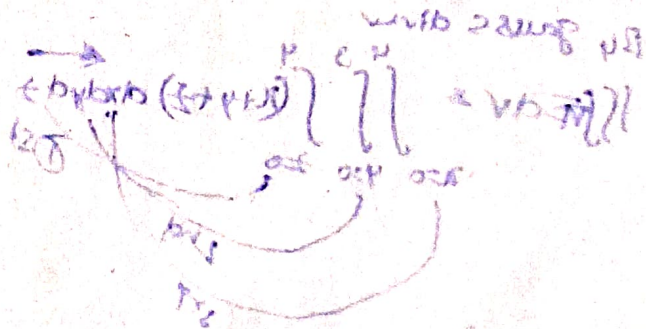
$$= \iint_D (a+a) dx dy = 2a \iint_D dx dy$$

$$= 2a \times \text{Area of circle} \Rightarrow 16\pi$$

$$= 2a \times \pi \cdot 2^2 = 8\pi a$$

$$16\pi = 8\pi a \Rightarrow a = 2$$

$$\vec{r} = \frac{y}{2}\vec{i} + \frac{y}{2}\vec{j} + x\vec{k} \Rightarrow \vec{r} = \frac{y}{2}(\vec{i} + \vec{j}) + x\vec{k}$$

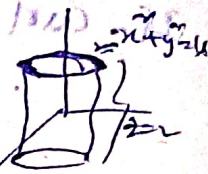


Gauss divergence theorem

$$\iint (\vec{F} \cdot \vec{n}) dS = \iiint (\nabla \cdot \vec{F}) dV$$

$$\iint F_1 dydz + F_2 dzdx + F_3 dxdy = \iiint \nabla \cdot \vec{F} dxdydz$$

Ex: $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$ Surf of cylinder $x^2 + y^2 = 16$ $-1 \leq z \leq 1$



By Gauss $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$, $\nabla \cdot \vec{F} = \frac{\partial}{\partial x} x + \frac{\partial}{\partial y} y + \frac{\partial}{\partial z} z = 3$

$$\nabla \cdot \vec{F} = 3 \quad \iint \vec{F} \cdot \vec{n} dS = \iiint \nabla \cdot \vec{F} dV$$

$$\iint \vec{F} \cdot \vec{n} dS = \iiint 3 dV \Rightarrow 3 \iiint dV \Rightarrow \text{Vol. of cylinder}$$

$$= 3 \times \pi \times r^2 \times h \Rightarrow 3 \times \pi \times 4^2 \times 2$$

Ex: $\iint_F xy dydz + yz dzdx + zx dxdy$ where S surf bounded by

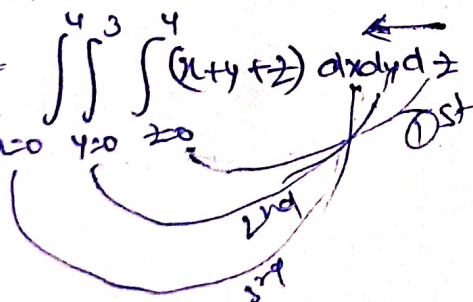
$$\begin{aligned} x &= 0 & x &= 4 \\ y &= 0 & z &= 0 \\ y &= 3 & z &= 4 \end{aligned}$$

By $\vec{F} = F_1\vec{i} + F_2\vec{j} + F_3\vec{k} \Rightarrow \vec{F} = xy\vec{i} + yz\vec{j} + zx\vec{k}$

$$\nabla \cdot \vec{F} = y + z + x$$

By Gauss div

$$\iiint \nabla \cdot \vec{F} dV = \int_{x=0}^4 \int_{y=0}^3 \int_{z=0}^4 (x+y+z) dxdydz$$



Multiple Integrals

Double Integral

when ever we have some fun or region
Some kind of rotation sym
but spreading expression x^2+y^2

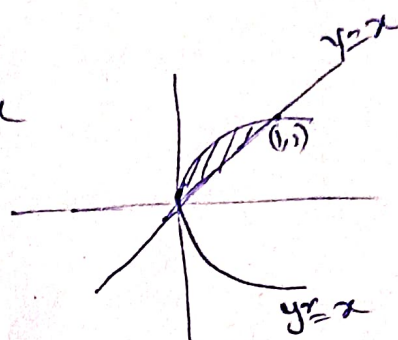
param form $x = r \cos \theta$, $y = r \sin \theta$, $dx dy = r dr d\theta$ $x^2+y^2 = r^2$

surf area of S in xy plane = $\iint_S r dr d\theta$

Area enclosed by curve $y^2 = x$ and $y = x$

surf area $\int_0^1 \int_{y^2}^y dx dy$

$$\Rightarrow \int_0^1 \int_{y^2}^y x dx dy = \int_0^1 \left(\frac{y^2 - x^2}{2} \right) dy$$



Triple Integral

$dv = dx dy dz$

$$\Rightarrow \int_0^1 \int_0^x \int_0^{x+y} x dx dy dz \Rightarrow \int_0^1 \int_0^x x dx dy$$

$$\Rightarrow \int_0^1 \int_0^x (x+y)x dx dy \Rightarrow \int_0^1 \int_0^x (x^2 + xy) dx dy$$

$$\Rightarrow \int_0^1 \left(x^2 y + \frac{xy^2}{2} \right) dy$$

$\iint dx dy \rightarrow$ surf area

$\iint f(x,y) dx dy \rightarrow$ vol which is projected $f(x,y)$

$\iiint dx dy dz \rightarrow$ vol of any irregular object

$\iiint f(x) dx dy dz =$ force applied on volume

$$\int [f(x)]^n f'(x) = \frac{f(x)^{n+1}}{n+1}$$

Polar form of triple Integral

Cylindrical co-ordinate

$x = r \cos \theta$, $y = r \sin \theta$, $z = z$, $x^2+y^2 = r^2$

$$\text{Volume} = \iiint r dr d\theta dz = \iiint r dr d\theta dz$$

Spherical co-ordinate

$x = r \sin \theta \cos \phi$, $y = r \sin \theta \sin \phi$, $z = r \cos \theta$

$$\text{Volume} = \iiint r^2 \sin \theta dr d\theta d\phi$$

Differential Equation (P.E)

Partial D.E

2D-Laplace Eqn

1. $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ - 1D Heat equation

3. $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \rightarrow$ 1D wave Eqn

5. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

2. $\frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$ - 2D Heat equation

4. $\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \rightarrow$ 2D wave Eqn

Order of D.E: Highest order derivatives

ex: $\frac{dy}{dx} + y^2 + e^x + \frac{dy}{dx^2} + \frac{d^4 y}{dx^4} = 0 \Rightarrow$ order = 4

ex: $\left(\frac{\partial u}{\partial y} \right)^3 + 3 \frac{\partial^2 u}{\partial x^2} + \frac{\partial^3 u}{\partial x \partial y^2} + \frac{\partial^3 u}{\partial y^3} = 0$ order = 3

Degree of D.E: Highest integral power of the highest order derivative

ex: $\left(\frac{dy}{dx} \right)^5 + \frac{dy}{dx^2} + x + \left(\frac{d^3 y}{dx^3} \right)^8 = 0$

Degree = 8

ex: $\frac{\partial u}{\partial x} + \left(\frac{\partial^2 u}{\partial x^2} \right)^3 + \left(\frac{\partial^3 u}{\partial x \partial y^2} \right)^5 + \left(\frac{\partial^3 u}{\partial y^3} \right)^9 = 0$

Degree = 9

ex: $y + e^x = \sin\left(\frac{dy}{dx}\right) \Rightarrow \sin^{-1}(y + e^x) = \frac{dy}{dx}$ order = 2, degree 1

ex: $\frac{d^3 y}{dx^3} + x \left(\frac{dy}{dx} \right)^{5/2} + x^2 y = 0$

square on both sides

$\Rightarrow \left(\frac{d^3 y}{dx^3} + x^2 y \right)^2 = \left(x \left(\frac{dy}{dx} \right)^{5/2} \right)^2$ order = 3, degree = 2

Linear D.E (L.D.E)

1. power b) degree $\rightarrow$ dependent var = 1

2. power b) degree $\rightarrow$ derivative var = 1

3. product $\rightarrow$ (D.V) x (derivative D.V) $\Rightarrow$ should not exist {i. $y^2 + x \frac{\partial y}{\partial x} + u \frac{\partial^2 u}{\partial x^2} = 0 \rightarrow$ Non-LDE

4. Transcendental fun should not be fun of D.V $\left\{ \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial y} + \sin(y) + u = 0 \rightarrow$ LDE

5. Non-Linear D.E: Not a LDE

ex: $x^3 + \frac{dy}{dx} + \frac{d^3 y}{dx^3} + \log(y) = 0$ - Non-LDE

Homogeneous D.E (H.D.E)

Every term contain Dependent (or) derivative variable

ex: $\left(\frac{dy}{dx} \right)^2 + \frac{d^3 y}{dx^3} + uxy^2 + y' + \log(y) = 0$ H.D.E

$e^x, \log(x), \text{Tri, Inverse Tri, Hyp, Inverse Hyp}$

Transcendental fun

8. Non-H.D.E: Not a H.D.E

If any term free from D.V or its derivative

ex: $x \frac{dy}{dx} + y + \frac{dy}{dx^2} + \sin x + 4x = 0 \rightarrow$ Non HDE

ex: $\frac{\partial^2 u}{\partial y^2} + 2x \frac{\partial u}{\partial y} + u^2 + 3 = 0$ Non Homogeneous part $\rightarrow$ Non-HDE

Homogeneous part

Methods of Finding G.S of $\frac{dy}{dx} = f(x, y)$

1. Variable Separable method (V.S.M)

IN. D.E $\frac{dy}{dx} = f(x, y) = h(x) \cdot g(y) \rightarrow$ V.S.M

ex: $\frac{dy}{dx} = (x+y-1)^2$ $x+y-1=z$

$\frac{dz}{dx} - 1 = z^2$ $1 + \frac{dz}{dx} = 0 = \frac{dz}{dx}$

$\frac{dz}{dx} = 1+z^2 \Rightarrow \int \frac{1}{1+z^2} dz = \int dx \Rightarrow \tan^{-1} z = x+c \Rightarrow \tan^{-1}(x+y-1) = x+c \Rightarrow y = 1-x + \tan(x+c)$

* V.S.M

$$\frac{dy}{dx} = h(x) \cdot g(y)$$

$$\frac{1}{g(y)} dy = h(x) dx$$

$$\int \frac{1}{g(y)} dy = \int h(x) dx + c \text{ gives G.S}$$

2. Homogeneous sol

$\frac{y}{x} = \frac{x}{y} = v \Rightarrow \frac{y}{x} = v =$

$y = vx$

$\frac{dy}{dx} = v + x \frac{dv}{dx}$

ex: $\sin\left(\frac{2x^2 - 2y}{y - 4x}\right)$

degree $n = 2 - 1 = 1$

ex: $f(x, y) = \frac{x^{3/2} + y^{3/2}}{2x^2 - y^2}$

degree $n = \frac{3}{2} - 2 = -1$

ex: $(x^2 + xy + y^2)dx - x^2 dy = 0$

$\frac{dy}{dx} = \frac{x^2 + xy + y^2}{x^2} \Rightarrow v + \frac{dv}{dx}x = \frac{x^2}{x^2} + \frac{xy}{x^2} + \frac{y^2}{x^2}$

let $\frac{y}{x} = v \Rightarrow y = vx$ $x + \frac{dv}{dx}x = 1 + v + v^2$

$\frac{dy}{dx} = v + x \frac{dv}{dx}$

$\int \frac{1}{1+v^2} dv = \int \frac{1}{x} dx \Rightarrow \tan^{-1}(v) = \log x + c \Rightarrow \tan^{-1}\left(\frac{y}{x}\right) = c + \log x$

Imp

Exact Method

$M(x, y)dx + N(x, y)dy = 0$ $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ Condition

$\int M(x, y) dx + \left[\text{collect the terms without } x \text{ in } N(x, y) \right] dy = c$

treating y as constant

ex: $\frac{dy}{dx} = \frac{-xy}{2+xy}$
 multiply both sides by $2+xy$ so we can do by homogen sol

5. Linear method — ^{character}_{told} (Max prob we can do by this method)

$$\frac{dy}{dx} + p(x) \cdot y = Q(x)$$

$$y(IF) = \int Q(x) \cdot (IF) dx + c$$

$$IF = e^{\int p(x) dx} \quad \text{— First find and substitute}$$

$$\text{ex: } \frac{dy}{dx} + 4y = 5 \Rightarrow \frac{dy}{dx} + \overset{4}{p(x)} y = \overset{5}{Q(x)}$$

$$\therefore IF = e^{\int 4 dx} = e^{4x}$$

$$y(e^{4x}) = \int 5 \cdot e^{4x} dx + c$$

$$y e^{4x} = \frac{5}{4} e^{4x} + c$$

$$\boxed{y = \frac{5}{4} + c e^{-4x}}$$

$$\frac{dx}{dy} + p(y)x = Q(y)$$

$$x \cdot (IF) = \int Q(y) (IF) dy + c$$

$$IF = e^{\int p(y) dy}$$

Second order D.E

Auxiliary Eqn (A.E)

Put $D = m$

$am^2 + bm + c$

$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

1. If roots are real and different

$$y_c = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

2. If roots are real and equal $m_1 = m_2 = m$

$$y_c = [c_1 + x c_2] e^{mx}$$

3. roots are complex $a \pm bi$

$$y_c = e^{ax} [c_1 \cos bx + c_2 \sin bx]$$

Model - 1

$F(D) y = e^{ax}$

Particular Integral (PI) = $\frac{1}{F(D)} e^{ax} \left[f(a) \neq 0 \right]$

D.Sol $y = y_c + y_p$

Ex: $(D^2 + 4D + 4)y = e^{-2x}$

sol $y_p = \frac{1}{D^2 + 4D + 4} e^{-2x}$

put $D = -2$
 $= \frac{1}{(D+2)(D+2)} e^{-2x}$

$= \frac{x^2}{2!} e^{-2x}$

$y_p = \frac{x^2}{2} e^{-2x}$

$D^2 + 4D + 4$

$D^2 + 2D + 2D + 4$

$D(D+2) + 2(D+2)$

$(D+2)(D+2)$ Imp

when becoming

zero $\frac{x^n}{n!}$

$n = \text{No. of becoming zero}$

Ex: $(D^2 - 16)y = \cosh 4x$

$y_p = \frac{1}{F(D)} \cosh 4x$

$= \frac{1}{D^2 - 16} \left\{ \frac{e^{4x} + e^{-4x}}{2} \right\}$

$= \frac{1}{2} \left\{ \frac{1}{D^2 - 16} e^{4x} + \frac{1}{D^2 - 16} e^{-4x} \right\}$

$= \frac{1}{2} \left\{ \frac{1}{(D+4)(D-4)} e^{4x} + \frac{1}{(D+4)(D-4)} e^{-4x} \right\}$

$= \frac{1}{2} \left\{ \frac{1}{4+4} \frac{x^1}{1!} e^{4x} + \frac{x^1}{1!} \frac{1}{(4-4)} e^{-4x} \right\}$

$y_p = \frac{1}{16} x e^{4x} - \frac{1}{16} x e^{-4x}$

Model - 2 $f(D) y = \sin ax$ (or) $\cos ax$

$y_p = \frac{1}{F(D)} \sin ax$

put $D^2 = -a^2$ if $f(-a^2) \neq 0$

If $f(-a^2) \neq 0$ then $y_p = \frac{x}{2} \int \sin ax \, dx$

If $f(-a^2)$ becoming zero

$y_p = \frac{x}{2} \int \cos ax \, dx$

$$\text{ex: } (D^2 + 5D + 6)y = \sin x + 1e^{-2x}$$

$$y_p = \frac{1}{D^2 + 5D + 6} \sin x + e^{-2x}$$

$$\begin{aligned} \text{D}^2 + 5\text{D} + 6 &= (\text{D} + 2)(\text{D} + 3) \\ \text{D}^2 + 5\text{D} + 6 &= (\text{D} + 2)(\text{D} + 3) \end{aligned}$$

$$= \frac{1}{-1^2 + 5D + 6} \sin x + \frac{1}{(D+3)(D+2)} e^{-2x}$$

$$= \frac{1}{5(D+1)} \times \frac{D-1}{D-1} \sin x + x e^{-2x}$$

$$= \frac{1}{5} \frac{D-1}{D^2 - 1} \sin x + x e^{-2x}$$

$$\text{Model: 3: } f(D)y = x^n$$

$$y_p = \frac{1}{f(D)} x^n$$

$$(1-D)^{-1} = 1 + D + D^2 + D^3 + \dots$$

$$(1+D)^{-1} = 1 - D + D^2 - D^3 + \dots$$

$$(1+D)^{-1} = 1 - D + D^2 - D^3 + \dots$$

$$\text{ex: } (D^2 + 4)y = x^3$$

$$A.E = m^2 + 4 = 0$$

$$m^2 = -4$$

$$m = \pm 2i$$

$$m = \pm 2i \Rightarrow a \pm bi$$

$$a=0, b=2$$

$$y_c = e^{ax} [C_1 \cos bx + C_2 \sin bx]$$

$$= e^0 [C_1 \cos 2x + C_2 \sin 2x]$$

$$y_c = C_1 \cos 2x + C_2 \sin 2x$$

$$y_p = \frac{1}{D^2 + 4} x^3$$

$$= \frac{1}{4 \left\{ 1 + \frac{D^2}{4} \right\}} x^3 \Rightarrow \frac{1}{4} \left(1 + \frac{D^2}{4} \right)^{-1} x^3$$

$$= \frac{1}{4} \left\{ 1 - \frac{D^2}{4} + \left(\frac{D^2}{4} \right)^2 \right\} x^3$$

$$= \frac{1}{4} \left\{ x^3 - \frac{D^2(x^3)}{4} - \frac{D^4(x^3)}{16} \right\}$$

$$= \frac{1}{4} \left\{ x^3 - \frac{6x}{4} \right\} \Rightarrow y_p = \frac{x^3}{4} - \frac{3x}{2}$$

$$y = y_c + y_p = C_1 \cos 2x + C_2 \sin 2x + \frac{x^3}{4} - \frac{3x}{2}$$

$$\begin{aligned} x^3 \\ D - 3x^2 \\ D^2 = 6x \\ D^3 = 6 \\ D^4 = 0 \end{aligned}$$

$$\boxed{\frac{x^3}{4} - \frac{3x}{2} = y_p}$$

Covering all models

$$(1+D)^{-1} = 1 - D + D^2 - D^3 \dots$$

Ex

$$(4D^2 + 4D + 1)y = e^{-x} + \cos x + x$$

Sol $AE = 4m^2 + 4m + 1 = 0$

$$4m^2 + 2m + 2m + 1 = 0$$

$$2m(2m+1) + 1(2m+1) = 0$$

$$(2m+1)(2m+1) = 0$$

$$m = -\frac{1}{2}, -\frac{1}{2} \text{ roots Equal}$$

$$Y_c = [C_1 + xC_2]e^{-\frac{1}{2}x}$$

$$D^2 = -1$$

$$= e^{-x} + \frac{4D(\cos x) + 3\cos x}{16(-1) - 9} + \left\{ x - 4D^2(x) - 4D(x) \right\} \Rightarrow e^{-x} + \frac{4(-\sin x) + 3\cos x}{-25} + (x-4)$$

$$Y = Y_c + Y_p = (C_1 + xC_2)e^{-\frac{x}{2}} + e^{-x} + \frac{4\sin x - 3\cos x}{25} + (x-4)$$

$$Y_p = \frac{1}{(2D+1)(2D+1)} e^{-x} + \cos x + x$$

put $D = -1$ put $D = -1$

$$= \frac{1}{(2D+1)^2} e^{-x} + \frac{1}{4D^2+4D+1} \cos x + \frac{1}{(1+4D^2+4D)} x$$

$$= \frac{1}{(2(-1)+1)^2} e^{-x} + \frac{1}{4(-1)+4D+1} \cos x + \frac{(1+(4D^2+4D))^{-1} x}{4D}$$

$$= e^{-x} + \frac{1}{-3+4D} \times \frac{4D+3}{4D+3} \cos x + \left\{ 1 - (4D^2+4D) + (4D^2+4D)^2 \dots \right\} x$$

$$= e^{-x} + \frac{4D+3}{(4D)^2-3^2} \cos x + \left\{ 1 - 4D^2 - 4D + 16D^4 + 16D^2 + 2 \times 4D \times D \right\} x$$

$$Y_p \Rightarrow e^{-x} + \frac{4\sin x - 3\cos x}{25} + (x-4)$$

Cauchy's Euler linear D.E

$$x = e^z \text{ or } \ln x = z$$

$$\left. \begin{aligned} xD &= 0 \\ x'D &= \theta(\theta-1) \end{aligned} \right\} \begin{aligned} \text{when } \theta &= \frac{d}{dz} \\ \theta &= \frac{d}{dx} \end{aligned}$$

Ex: $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = 0$, $x \geq 1$ with Initial Cond $y(x=1) = 6$, $\frac{dy}{dx} \Big|_{x=1} = 2$, then $y(x=2) = ?$

Sol: $(x^2 D^2 + xD - 1)y = 0$ $xD = 0$
 $\{ \theta(\theta-1) + \theta - 1 \} y = 0$ $x'D = \theta(\theta-1)$
 from this

$$\Rightarrow y' = C_1 + C_2 \left(\frac{1}{x^2} \right)$$

$$y(x=1) = 6 \Rightarrow y = C_1 x + \frac{C_2}{x} = 6 \rightarrow \textcircled{1}$$

$$y'(1) = 2 \Rightarrow C_1 + C_2 \left(\frac{-1}{1^2} \right) = 2$$

$$C_1 - C_2 = 2 \rightarrow \textcircled{2}$$

Solve $\textcircled{1}$ & $\textcircled{2}$ $C_1 = 4, C_2 = 2$

Sol: $y = y(x) = 4x + \frac{2}{x}$

$$y(x) = 4(x) + \frac{2}{2} = 9$$

Ex: $x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + 2y = 4$

Sol: $\{ \theta(\theta-1) - 2\theta + 2 \} y = 4$

complementary fun c.f

At $m^2 - 3m + 2 \leftarrow (\theta^2 - \theta - 2\theta + 2)$
 $m = 1, 2$ $\theta^2 - 3\theta + 2$

$$y_c = C_1 e^x + C_2 e^{2x}$$

$$y_c = C_1 x + C_2 x^2$$

particular PI

$$4 \Rightarrow 4e^{0x}$$

$$y_p = \frac{1}{f(\theta)} e^{ax} \quad \theta = a$$

$$y_p = \frac{1}{\theta^2 - 3\theta + 2} e^{0x}$$

$$= \frac{1}{(\theta-2)(\theta-1)} e^{0x} \Rightarrow \frac{1}{2}$$

Complex Variables

$$z = x + iy \quad (\text{Cartesian form}) \Rightarrow z = r(\cos\theta + i\sin\theta)$$

$$z = r(\cos\theta + i\sin\theta)$$

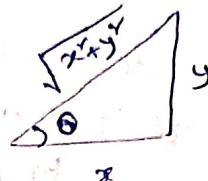
$$z = re^{i\theta}$$

$$z = re^{i\theta} \quad -\pi < \theta \leq \pi \quad \text{Principal value}$$

$$z = re^{i(\theta + 2n\pi)} \quad \text{General value}$$

Used to find $\arg z$ of complex num

$$r = \sqrt{x^2 + y^2}$$



Euler's formula

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$e^{-i\theta} = \cos\theta - i\sin\theta$$

1. If z lie in first quadrant then

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

2. If z lie in 2nd quadrant then

$$\theta = \pi - \tan^{-1}\left(\frac{y}{|x|}\right)$$

3. If z lie in 3rd quadrant then

$$\theta = -\pi + \tan^{-1}\left\{\frac{|y|}{|x|}\right\}$$

4. If z lie in 4th quadrant then

$$\theta = -\tan^{-1}\left[\frac{|y|}{x}\right]$$

$$\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$$

$$\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$$

3. Let $w^4 = 16i$ which of following can not be value of w a) $2e^{i\frac{2\pi}{8}}$ b) $2e^{i\frac{\pi}{8}}$ c) $2e^{i\frac{5\pi}{8}}$ d) $2e^{i\frac{9\pi}{8}}$

Sg Some times $i = i$

$$w^4 = 16i$$

$$w = (16i)^{1/4}$$

$$= 16^{1/4} i^{1/4}$$

$$w = 2 \left(0 + i(1)\right)^{1/4}$$

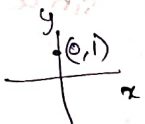
$$r = \sqrt{x^2 + y^2} = \sqrt{0^2 + 1^2} = 1$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$= \tan^{-1}\left(\frac{1}{0}\right)$$

We can take 1st or 2nd

$$\theta = \pi/2$$



$$w = 2 \left\{ re^{i(\theta + 2n\pi)} \right\}^{1/4}$$

$$w = 2 \left\{ 1 e^{i(\frac{\pi}{2} + 2n\pi)} \right\}^{1/4}$$

$$\text{for } n=0 \quad w = 2 \left(e^{i\pi/2} \right)^{1/4} = 2e^{i\pi/8}$$

$$\text{for } n=1 \quad w = 2 \left\{ e^{i(\frac{\pi}{2} + 2\pi)} \right\}^{1/4} = 2e^{i5\pi/8}$$

$$\text{for } n=2 \quad w = 2 \left\{ e^{i(\frac{\pi}{2} + 4\pi)} \right\}^{1/4} = 2e^{i9\pi/8}$$

$$n=3 \quad w = 2 \left\{ e^{i(\frac{\pi}{2} + 6\pi)} \right\}^{1/4} = 2e^{i13\pi/8}$$

Here we need so we need see find 4 roots

(or) another method option check method

$$2e^{i\pi/8} = w^4$$

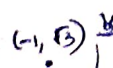
$$(2e^{i\pi/8})^4 = 16e^{i\pi/2} = 16 \left[\cos\frac{\pi}{2} + i\sin\frac{\pi}{2} \right] = 16(0 + i(1)) = 16i$$

4. Find principal & general value $\ln(-1 + i\sqrt{3})$

$$\ln(-1 + i\sqrt{3}) = \ln(re^{i\theta})$$

Cartesian coordinate form $\uparrow$ polar co-ordinate form

$$r = \sqrt{x^2 + y^2} = \sqrt{(-1)^2 + (\sqrt{3})^2} = 2$$



$$\theta = \pi - \tan^{-1}\left(\frac{y}{|x|}\right) = \pi - \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = \frac{2\pi}{3}$$

$$\text{principal val} = \ln(-1 + i\sqrt{3}) = \ln(2e^{i\frac{2\pi}{3}})$$

$$\Rightarrow \ln(2e^{i\frac{2\pi}{3}}) = \ln 2 + \ln e^{i\frac{2\pi}{3}} = \ln 2 + i\left(\frac{2\pi}{3}\right)$$

$$\text{General val } \ln(-1 + i\sqrt{3}) = \ln(re^{i(\theta + 2n\pi)})$$

$$= \ln \left\{ 2e^{i\left(\frac{2\pi}{3} + 2n\pi\right)} \right\} = \ln 2 + i\left(\frac{2\pi}{3} + 2n\pi\right)$$

Cauchy-Riemann Eqn (CR Eqn)

$$\boxed{\begin{aligned} u_x &= v_y & \left\{ \frac{\partial u}{\partial x} = u_x, \frac{\partial v}{\partial y} = v_y \right\} \\ u_y &= -v_x \end{aligned}}$$

3. A fun $f(z) = xz$ is

- a) differentiable at $(0,0)$
b) analytic at $(0,0)$

- c) differentiable at $(1,-2)$
d) analytic everywhere

Sm Given $f(z) = xz = x(x+iy) = \underbrace{x^2}_u + i \underbrace{xy}_v$

$$\frac{\partial u}{\partial x} \Rightarrow u_x = 2x, \quad u_y = 0, \quad v_x = y, \quad v_y = x$$

$$\begin{aligned} u_y &\neq -v_x & \text{CR Eqn} \\ u_x &\neq v_y & \text{not satisfied} \end{aligned}$$

If eqn not satisfied go by points

$$\text{at } (0,0) \quad u_x = 2x = 2(0) = 0, \quad u_y = 0, \quad v_x = y = 0, \quad v_y = x = 0$$

$$\begin{aligned} u_x &= v_y \\ u_y &= -v_x \end{aligned}$$

So finite at $(0,0)$ and continuous at $(0,0)$

• CR eqn satisfied at $(0,0)$

$f(z)$ is differentiable at $(0,0)$

$f(z)$ not analytic at $(0,0)$ since CR eqn not satisfied in neighbourhood of $(0,0)$

Construction of Analytic fun (Model-I)

$$f(z) = u + iv$$

$\downarrow$ Real part
 $\downarrow$ Imaginary part
 A.K.A

By total derivative definition

$$dv = V_x dx + V_y dy$$

$$dv = -u_y dx + u_x dy$$

Integration on both sides

$$\int dv = \int -u_y dx + \int u_x dy$$

$$v = -\int u_y dx + \int (\text{terms of } u_x \text{ do not contain } x) dy$$

$y = \text{const}$

Similarly real part u can find if imaginary part v given

3) Analytic fun of complex variable $z = x + iy$ in $f(z) = x^2 + y^2 + i\psi(x, y)$ then imaginary part of $f(z)$ at $z = 1 + i$ is —

So $f(z) = x^2 + y^2 + i\psi(x, y)$ ($\psi = v$ here)

$$\text{Let } u = x^2 + y^2 \Rightarrow u_y = 2y, u_x = 2x$$

$$d\psi = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy \quad (\text{T.D of } \psi)$$

$$d\psi = -u_y dx + u_x dy$$

$$= -2y dx + 2x dy$$

$$\int d\psi = \int -2y dx + \int 2x dy$$

$$y = \text{const} \quad \downarrow \text{term doesn't contain } x$$

$$\text{at } z = 1 + i \quad (1, 1)$$

$$\psi = 2xy$$

$$\psi(1, 1) = 2 \times 1 \times 1$$

$$= 2 \times 1 \times 1 = 2$$

Model II If u given find $f(z) = u + iv$ without finding v directly $f(z)$

Consider

$$f'(z) = u_x + i v_x$$

$$f'(z) = u_x - i u_y$$

Putting $x = z$ and $y = 0$ in $f'(z)$

$$f'(z) = g(z) \text{ say}$$

$$\int f'(z) dz = \int g(z) dz$$

$$f(z) = \int g(z) dz + C$$

C.R Eqn
 $u_x = v_y$
 $u_y = -v_x$

1. If $u = (x-1)^3 - 3xy^2 + 3y^3$ real part of analytic fun $f(z) = u + iv$ then $f(z)$ is —

$$\Rightarrow u = (x-1)^3 - 3xy^2 + 3y^3$$

$$f'(z) = u_x + i v_x \Rightarrow u_x + i(-u_y)$$

$$= u_x - i u_y \quad u_x = \frac{\partial u}{\partial x} = 3(x-1)^2 - 3y^2$$

$$= 3(x-1)^2 - 3y^2 - i(-6xy + 6y^2) \quad u_y = -6xy + 6y^2$$

$$\text{put } x = z, y = 0 \quad f'(z) = 3(z-1)^2 - 0 - 0$$

$$\int f'(z) dz = \int 3(z-1)^2 dz$$

$$f(z) = (z-1)^3 + C$$

2. If $v = y^3 + 2y - 3x^2y$ where $z = x + iy$

then $u = ?$ a) $u = 2x - x^3 + 3xy^2 + C$ b) $u = 3x - x^3 + 3xy^2 + C$ c) $f(z) = 2z - z^3 + C$

d) $f(z) = 2i(z) + z^3 + C$

So $v = y^3 + 2y - 3x^2y \quad v_x = -6xy, v_y = 3y^2 + 2 + 3x^2$

option verification let $u = 2x - x^3 - 3xy^2 + C$ then $u_x = 2 - 3x^2 - 3y^2, u_y = -6xy$

so $u_x = v_y$ & $u_y = -v_x$ (C.R Eqn satisfied so option A correct)

Cauchy goursat theorem (Cauchy Integral theorem) should be used.

i) If one or more singular points lie inside the curve then Cauchy's integral formula should be used

ii) Circle is $|z - z_0| = 1$

$$|(x+iy) - (x_0+iy_0)| = 1$$

$$|(x-x_0) + i(y-y_0)| = 1$$

$$\Rightarrow \sqrt{(x-x_0)^2 + (y-y_0)^2} = 1$$

$$\Rightarrow (x-x_0)^2 + (y-y_0)^2 = 1$$

center $\Rightarrow (x_0, y_0)$ real can be

$\Rightarrow (x_0 + iy_0) \neq z$ represented in complex

iii) unit circle with centre

at origin is $|z| = 1$

$$|x+iy| = 1 \Rightarrow \sqrt{x^2 + y^2} = 1 \Rightarrow x^2 + y^2 = 1$$

Problems

1) If C is a closed path in the complex

plane then $\oint_C e^{-z^2} dz = 0$

2) e^{-z^2} is analytic everywhere

e^{-z^2} has no singularities. By Cauchy's Integral theorem

$$\oint_C e^{-z^2} dz = 0$$

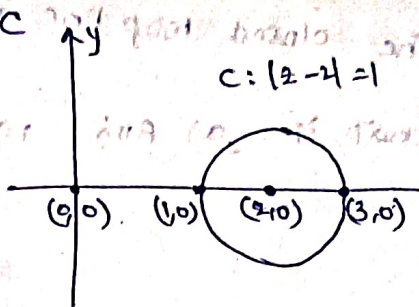
3) The value of $\oint_C \frac{z^2 + \cos z}{(z-4)^3(z+2)} dz$ along a curve $C: |z-2| \leq 1$ is

Q1) $z = -2 = -2 + i(0) = (-2, 0)$ lies outside C

$$(z-4)^3(z+2) = 0$$

$$z+2 = 0 / (z-4)^3$$

$$z+2=0 \Rightarrow \boxed{z=-2}$$



$$(z-4)^3 = 0$$

$$z-4 = 0$$

$$\boxed{z=4} = 4 + i(0) = (4, 0)$$

lie outside C

By Cauchy's Integral theorem we have

$$\oint \frac{z^7 \cos z}{(z-4)^3(z+2)} dz = 0$$

all points are outside the curve so we should use the Cauchy Integral theorem

Circle Eqn for Complex Variable

$$|z - z_0| = r$$

↓ ↓
centre radius

$$|z - 2| = 1$$

↓
(2,0) → centre

Q2) C is a closed path in the z plane given by $|z|=3$ - the value of the Integral $\oint_C \frac{z^7 - z + 4j}{z+2j} dz$ is

- a) $-4\pi(1+2j)$ c) $-4\pi(3+2j)$
b) $2\pi(3-2j)$ d) $4\pi(1-2j)$

$$\oint \frac{(z^7 - z + 4j)}{(z - (-2j))} dz = 2\pi j f(-2j)$$

↓
 z_0

Sub $z = -2j$ in $f(z)$

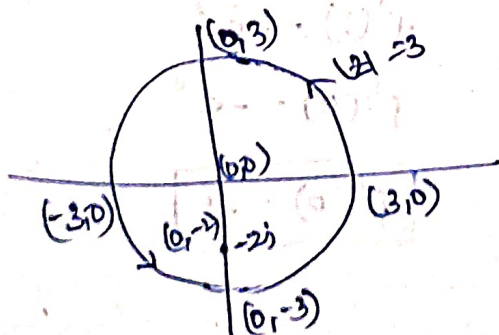
$$= 2\pi j \{ (-2j)^7 - (-2j) + 4j \}$$

$$= 2\pi j \{ 4j^7 + 6j \} \Rightarrow 2\pi j \{ -4 + 6j \}$$

$$= 2\pi(-4j - 6) \Rightarrow -4\pi(2j + 3) //$$

$z + 2j = 0 \Rightarrow z = -2j = (0, -2)$ lies inside $|z|=3$.

So we use Cauchy Integral formula



4) The closed loop line integral $\oint \frac{z^3 + 2z + 8}{z+2} dz$ evaluate counter

clockwise in a) $8\pi i$ b) $-8\pi i$ c) $-4\pi i$ d) $4\pi i$



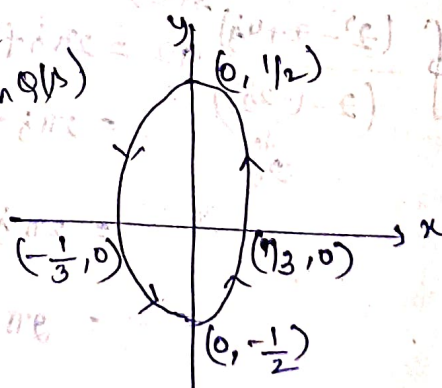
5. The value of $\oint \frac{\sin(\frac{\pi z}{2})}{(z-1)} dz$ where C is $|z-2|=4$ is $2\pi i$

6. The value of $\oint_C \frac{1}{z^2 e^z} dz$ where C is $qx^2 + 4y^2 = 1$ is a) $2\pi i$ b) 0 c) πi d) 2π

Q9 $qx^2 + 4y^2 = 1$

$$\frac{x^2}{(\frac{1}{3})^2} + \frac{y^2}{(\frac{1}{2})^2} = 1$$

(do not given in Q/s)



$$\oint \frac{dz}{z^2 e^z} = \oint \frac{e^{-z}}{(z-0)^{1+1}} = 2\pi i \frac{f'(0)}{1!}$$

$$= -2\pi i$$

$$f(z) = e^{-z}$$

$$f'(z) = -e^{-z}$$

$$f'(0) = -e^0$$

$$f'(0) = -1$$

7) The value of $\oint \frac{e^{2z}}{(z+1)^4}$ where C is the circle $|z|=2$ is $\frac{8}{3}\pi i e^{-2}$

$$\frac{e^{2z} \rightarrow f(z)}{(z - (-1))^{3+1}} = \frac{2\pi i f^{(3)}(z_0)}{3!}$$

$$= 2\pi i \frac{8e^{2z}}{3 \times 2 \times 1}$$

$$= \frac{8}{3}\pi i e^{-2}$$

$$(z+1)^4 = 0$$

$$z+1=0$$

$$(z = -1)$$

lies inside circle

$$\frac{f(z)}{(z - z_0)^{n+1}} = \frac{2\pi i f^{(n)}(z_0)}{n!} \quad |z|=2$$

$$f(z) = e^{2z}$$

$$f'(z) = 2e^{2z}$$

$$f'' = 4e^{2z}$$



$$\Rightarrow \boxed{f = 8e^{2z}} \Rightarrow 8e^{-2}$$

8) If C is the boundary of $4x^2 + 9y^2 = 36$ then $\oint \frac{\cos z + ie^{iz}}{(z + \frac{\pi}{2})^3} dz$ is $\frac{\pi e^{-\pi/2}}{2}$

9) Let $(-1-j), (3-j), (3+j)$ and $(-1+j)$ be the vertices of a rectangle C' in the complex plane. Assuming that C' is traversed in counter clockwise direction the value of contour integral $\oint \frac{dz}{z^2(z-4)}$ is

$z=0$ lies inside rectangle $z=4$ lies outside rectangle

$z=0 \Rightarrow (z=0) = 0+i(0) = (0,0)$ lies inside $\square$
 $z=4 \Rightarrow (z=4) = 4+i(0) = (4,0)$ lies outside $\square$

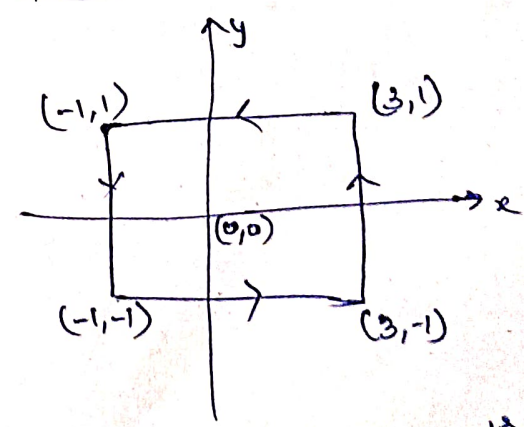
$$\oint \frac{dz}{z^2(z-4)} = \oint \frac{\frac{1}{z^2}}{z-4} = \frac{f'(z)}{(z-0)^{2+1}}$$

$$\Rightarrow 2\pi i \frac{f'(0)}{1!} \Rightarrow 2\pi i \left(-\frac{1}{16}\right)$$

$$\Rightarrow -\frac{\pi i}{8}$$

$$f'(z) = \frac{-1}{(z-4)^2}$$

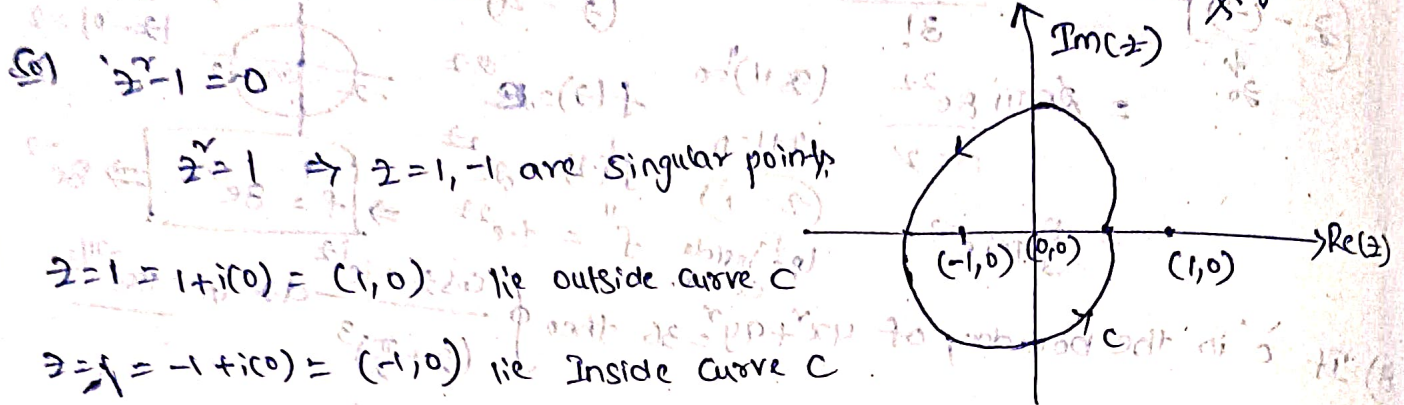
$$f'(0) = \frac{-1}{(0-4)^2} = -\frac{1}{16}$$



Outside lie Numerator
 Inside lie denominator

10. A simple closed path c in the complex plane is shown in fig

If $\oint \frac{z^2}{(z^2-1)} dz = -i\pi A$, (where $i^2 = -1$ then value of A is 0.5)



$$\oint_C \frac{z^2}{(z^2-1)} dz = \oint \frac{z^2}{(z+1)(z-1)} dz$$

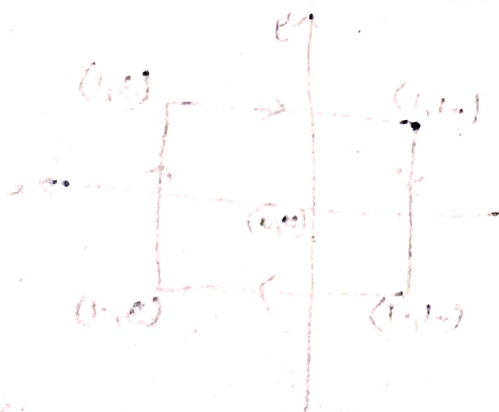
o/side Numerator
I/side denominator

$$= \oint \frac{\frac{z^2}{(z-1)}}{\underset{z_0}{[z-(-1)]}} dz \Rightarrow 2\pi i \frac{f'(z)}{1} \Rightarrow 2\pi i \frac{z^{-1}}{(-1-1)} \Rightarrow \frac{2\pi i}{-2} \Rightarrow -\frac{\pi i}{1}$$

$\therefore \Rightarrow -\frac{\pi i}{2} = -i\pi A$ By Comparing $A = \frac{1}{2} = 0.5$

11) The value of $\oint_C \frac{\cos(\pi z^3)}{(z-1)(z-3)} dz$ where C is $|z|=2$ is $-\pi i$

So



$(z-1)(z-3) = (z-1)(z-3)$

$(1,1) (1,3) (-1,1) (-1,3)$

$\frac{1}{(z-1)(z-3)} = \frac{A}{z-1} + \frac{B}{z-3}$

$\frac{1}{(z-1)(z-3)} = \frac{A(z-3)}{(z-1)(z-3)} + \frac{B(z-1)}{(z-1)(z-3)}$

$\left(\frac{1}{z}\right) \ln z = \frac{(z)^1 \ln z}{1!}$

$\frac{1}{z} = \frac{1}{z}$

Q1) The contour C given below is on the complex plane $z = x + jy$

where $j = \sqrt{-1}$. The value of $\frac{1}{\pi j} \oint_C \frac{dz}{(z^2-1)}$ is.

$$\Rightarrow \frac{1}{\pi j} \oint_C \frac{dz}{(z^2-1)} = \frac{1}{\pi j}$$

$$\frac{1}{(z^2-1)} = \frac{1}{(z+1)(z-1)} = \left\{ \frac{(z+1)(z-1)}{(z+1)(z-1)} \right\} \frac{1}{2}$$

$$\Rightarrow \frac{1}{2} \left\{ \frac{1}{(z-1)} - \frac{1}{(z+1)} \right\}$$

$$\oint_C \frac{dz}{z^2-1} = \frac{1}{2} \left\{ \oint_C \frac{1}{z-1} dz - \oint_C \frac{1}{z+1} dz \right\}$$

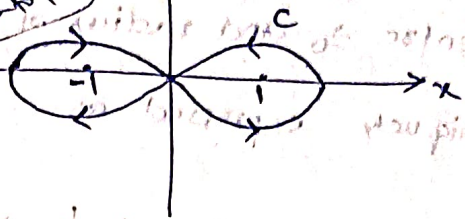
$$\Rightarrow \frac{1}{2} \left\{ \oint_C \frac{1}{z-1} dz - \oint_C \frac{1}{z-(-1)} dz \right\} \Rightarrow \frac{1}{2} \left\{ 2\pi i f(1) + 2\pi i f(-1) \right\}$$

Anti Clockwise Clockwise

$$\Rightarrow \frac{1}{2} \left\{ 2\pi i (1) + 2\pi i (1) \right\} \Rightarrow 2\pi i$$

$$\frac{1}{\pi j} \oint_C \frac{dz}{(z^2-1)} = \frac{1}{\pi j} (2\pi i) \Rightarrow 2 //$$

addition of subtraction of denominator should write in two terms such that numerators should not change



$z^2-1=0 \Rightarrow z=1, -1$ are singular points

Both singular points in curve C

due to clockwise multiply by -1

Singular point which is lying outside is taken in Numerator

Suppose If the two singular points lie

Inside the Curve then split the fun in partial fractions